

02-25-2025

Review:

HW1P1:

$$1. C_1 = y_1 - \rho a_2 \Rightarrow a_2 = \frac{y_1 - c_1}{\rho}$$

$$C_2 = a_2 + y_2 - \rho a_3 \Rightarrow y_2 + \frac{y_1 - c_1}{\rho} - \rho a_3$$

$$\Rightarrow a_3 = \frac{y_2 - c_2}{\rho} + \frac{y_1 - c_1}{\rho^2}$$

$$C_3 = y_3 + a_3 - \rho a_4 \Rightarrow y_3 + \frac{y_1 - c_1}{\rho^2} + \frac{y_2 - c_2}{\rho} - \rho a_4$$

$\Rightarrow$

$$C_T = y_T + \frac{y_{T-1} - c_{T-1}}{\rho} + \dots + \frac{y_1 - c_1}{\rho^{T-1}} - \rho a_{T+1}$$

$$C_1 + C_2 \rho + \dots + C_T \rho^{T-1} = y_1 + y_2 \rho + \dots + y_T \rho^{T-1} \quad 0$$

$$2. \sum_{t=1}^T \rho^{t-1} u(c_t)$$

s.t.

$$\sum_{t=1}^T \rho^{t-1} c_t = \sum_{t=1}^N \rho^{t-1} y_t \Rightarrow u'(c_t) = \frac{\partial}{\rho} u'(c_{t+1})$$

$$\Rightarrow c_{t+1} = \frac{\partial}{\rho} c_t$$

$$2: q = \frac{1}{1+r} \Rightarrow c_{t+1} = \theta(1+r) c_t$$

$$q = \frac{\theta}{1+r} \Rightarrow c_{t+1} = (1+r) c_t$$

$$1) r=0 \Rightarrow q = \frac{1}{1+r} = 1 \Rightarrow \sum c_t = N_y$$

$$c_{t+1} = \theta c_t \Rightarrow c_t = \theta^{t-1} c_1$$

$$c_1 + \theta c_1 + \dots + \theta^{T-1} c_1 = N_y$$

$$c_1 = \frac{1-\theta}{1-\theta^T} N_y, c_t = \theta^{t-1} \frac{1-\theta}{1-\theta^T} N_y$$

(decreasing)

$$2) q = \theta$$

$$c_{t+1} = c_t = c$$

$$\sum_{t=1}^T \theta^{t-1} c_t = y \sum_{t=1}^T \theta^{t-1} \Rightarrow c = y \frac{1-\theta^T}{1-\theta} \text{ (flat } c)$$

HW1: P2

$$c_t = E[c_t] \Rightarrow \text{After } 40 \Rightarrow c_t = c_{t+1} = \frac{a_{41}}{20}$$

$$t=40 \Rightarrow c_t = c_{t+1} = a_{40} + y_{40} - a_{41} = \frac{a_{41}}{20}$$

$$\Rightarrow a_{41} = \left(\frac{20}{21}\right)(a_{40} + y_{40}) \Rightarrow c_{40} = \frac{1}{21} a_{40} + \frac{1}{21} y_{40}$$

$$t=39 \Rightarrow a_{39} + y_{39} - a_{40} = E_{39} \left[ \frac{1}{21} a_{40} + \frac{1}{21} (y_{39} + \varepsilon_{40}) \right]$$

$$\Rightarrow a_{40} = \frac{21}{22} a_{39} + \frac{20}{22} y_{39} \Rightarrow c_{39} = \frac{1}{22} a_{39} + \frac{2}{22} y_{39}$$

HW1: P2:

$$a_{t+1} = \frac{20 + (41-t)}{20 + (41-t) + 1} a_t + \frac{20}{20 + (41-t) + 1} y_t$$

$$c_t = \frac{1}{20 + (41-t)} a_t + \frac{41-t}{20 + (41-t)} y_t$$

$$\Delta c_{t+1} = \frac{40-t}{60-t} \varepsilon_{t+1}$$

HW2: P1:

$$1) E_c = \exp \left\{ E \log c + \frac{1}{2} \text{var} \log c \right\} = e^0 = 1$$

$$\frac{\mu}{1-\alpha} \quad \frac{\sigma_\varepsilon^2}{1-\alpha^2}$$

$$2) V(\lambda) = E_{-1} \sum_{t=0}^{\infty} \beta^t \frac{(c_t + (1+\lambda))^{1-\gamma}}{1-\gamma}$$

by LIE

$$\Rightarrow \frac{(1+\lambda)^{1-\gamma}}{(1-\gamma)(1-\beta)} E_{-1} c^{1-\gamma}$$

exp doesn't change

$$= \frac{(1+\lambda)^{1-\gamma}}{(1-\gamma)(1-\beta)} \exp \left\{ \frac{\mu(1-\gamma)}{1-\alpha} + \frac{1}{2} \frac{\sigma_\varepsilon^2 (1-\gamma)^2}{1-\alpha^2} \right\}$$

$$V_0 = \sum_{t=0}^{\infty} \beta^t \frac{1}{1-\sigma} = \frac{1}{(1-\beta)(1-\gamma)}$$

HW2: P1:

$$2) V_0 = \sum_{t=0}^{\infty} \beta^t \frac{1^{1-\gamma}}{1-\gamma} = \frac{1}{(1-\beta)(1-\gamma)}$$

$$V(\lambda) = V_0$$

$$1+\lambda = \exp \left\{ \frac{\gamma \sigma_\varepsilon^2}{2(1-\alpha^2)} \right\}$$

↗  
comp for uncertainty

$$D) c_{t+1|t} \sim N \left( \mu + \alpha \log c_t, \sigma_\varepsilon^2 \right)$$

$$E_t c_{t+1} = \exp \left\{ -\gamma \mu - \gamma \alpha \log c_t + \frac{1}{2} \gamma^2 \sigma_\varepsilon^2 \right\}$$
$$\log c_t = -\frac{1}{\gamma} \frac{1}{1-\alpha} \log(\beta(1+r)) + \frac{\mu}{1-\alpha} - \frac{1}{2} \frac{\gamma}{1-\alpha} \sigma_\varepsilon^2$$

## Practice Midterm:

1) find log-linear

$L_{\text{palmost}}$  identical to  $\text{HWZ}(1)$

2) 4 states:

4 States:

$s, u$     $m, u$     $n$  (relationship skill)  
 $s, e$     $m, e$     $w$  ( $c, v, h, n$ )  $\cdot E_w$

characteristic of job

shock

nomax

$$\begin{aligned}
 V^{s,u}(h,n,u) &= \log(\text{cons}) + \beta(1-\gamma^m)(1-\gamma^u) \\
 &V^{s,v}(h',n,u+1) + \beta(1-\gamma^m)\gamma^u \sum_c \sum_v \sum_{\epsilon'_\omega} \prod \psi^\omega \\
 &\cdot V^{s,e}(h',n,c,v,\epsilon'_\omega) + \beta\gamma^m(1-\gamma^u) \sum \psi^m \\
 &\cdot V^{m,u}(h',n,u+1,\epsilon'_m) + \beta\gamma^m\gamma^u \sum_c \sum_v \sum_{\epsilon'_m} \sum_{\epsilon'_\omega} \prod \tilde{\psi}^m \psi^\omega \\
 &\cdot V^{m,e}(h',n,c,v,\epsilon'_\omega,\epsilon'_m)
 \end{aligned}$$

$$\text{cons} = \beta(u); \quad h' = h - \delta$$

$$V^{s,e}(h, n, c, \gamma, \varepsilon_\omega) = \log(\text{cons}) + \beta(1-\gamma^m) \sum_{\varepsilon'_\omega} \psi^\omega \\ \cdot \max \{ V^{s,e}(h', n, c, \gamma, \varepsilon'_\omega), V^{s,u}(h', n, 0) \} + \\ \beta \gamma^m \sum_{\varepsilon'_n} \sum_{\varepsilon'_\omega} \psi^\omega \psi^m \cdot \max \{ V^{m,e}(h', n, c, \gamma, \varepsilon'_\omega, \varepsilon'_m), \\ V^{m,u}(h', n, 0, \varepsilon'_m) \} \quad \text{cons} = \omega, h' = h + \delta$$

$$V^{m,u}(h, n, u, \varepsilon_m) = \log(\text{cons}) \cdot \varepsilon_m + \beta(1-\gamma^u) \sum_{\varepsilon'_m} \psi^m \\ \cdot \max \{ V^{m,u}(h', n, u+1, \varepsilon'_m), V^{s,u}(h', n, u+1) \} \\ + \beta \gamma^u \sum_c \sum_\gamma \sum_{\varepsilon'_\omega} \sum_{\varepsilon'_m} \prod \psi^m \psi^\omega \cdot \max \{ V^{m,e}(h', n, c, \gamma, \varepsilon'_\omega, \varepsilon'_m), \\ V^{s,e}(h', n, c, \gamma, \varepsilon'_\omega) \}$$

$$V^{m,e}(h, n, c, \gamma, \varepsilon_\omega, \varepsilon_m) = \log(\text{cons}) \varepsilon_m + \\ \beta \sum_{\varepsilon'_m} \sum_{\varepsilon'_\omega} \psi^m \psi^\omega \max \{ V^{m,e}(h', n, c, \gamma, \varepsilon'_\omega, \varepsilon'_m), \\ V^{s,e}(h', n, c, \gamma, \varepsilon'_\omega), V^{m,u}(h', n, 0, \varepsilon'_m), \\ V^{s,u}(h', n, 0) \}$$

$$h' = h + \delta$$

$$\text{cons} = \omega$$

HWZ: 2:

$$2: v_t^p = (1-\beta)c_t^p + \beta v_{t+1}^p$$

$$c = \bar{c} \Rightarrow v_0 = \bar{c}$$

$$v_0^p = 1 - \beta \bar{c}^p + \beta v_1^p = (1-\beta)\bar{c}^p + \beta \overbrace{((1-\beta)\bar{c}^p + \beta v_2^p)}^{v_1^p}$$

$$\dots = (1-\beta)\bar{c}^p(1 + \beta + \beta^2 + \dots) = \bar{c}^p \quad \text{w/ uncert}$$

$$3: U_t = [E_t v_t^\alpha]^\frac{1}{\alpha} \Rightarrow U_t(1+\eta) = [E_t v_t^\alpha (1+\eta)^\alpha]^\frac{1}{\alpha}$$

$$U_0(1+\eta) = \bar{c} \Rightarrow \text{no uncert}$$

$$1+\eta = \frac{\bar{c}}{U_0}$$

$$4: \begin{cases} \frac{1}{2} c_\ell, \dots, c_\ell \leftarrow v_{0\ell} \\ \frac{1}{2} c_h, \dots, c_h \leftarrow v_{0h} \end{cases}$$

$$U_0 = \left[ \frac{1}{2} c_\ell^\alpha + \frac{1}{2} c_h^\alpha \right]^\frac{1}{\alpha}$$

$$1+\eta = \frac{\bar{c}}{\left[ \frac{1}{2} c_\ell^\alpha + \frac{1}{2} c_h^\alpha \right]^\frac{1}{\alpha}}$$

$$5: \begin{cases} \frac{1}{2} c_\ell \\ \frac{1}{2} c_h \end{cases}$$

$$U_0 = [E_0 v_0^\alpha]^\frac{1}{\alpha} \Rightarrow \frac{1}{2} \left[ [(1-\beta)c_\ell^p + \beta v_0^p]^\frac{\alpha}{p} + [(1-\beta)c_h^p + \beta v_0^p]^\frac{\alpha}{p} \right]^\frac{1}{\alpha}$$

HW3: P1:

$$1) V_t(k_t, z_t, x_t) = \max \{ V_t^{fd}(k_t, z_t, x_t), V_t^w(k_t, z_t, x_t) \}$$

$$V_t^w(k_t, z_t, x_t) = \max_{c_t, k_{t+1}, l_t} \{ u(c_t, l_t) + \beta$$

$$\sum_{x_{t+1}} \psi(x_{t+1} | x_t) \times \left[ (1-d) \sum_{z_{t+1}} \pi(z_{t+1} | z_t) \right.$$

$$\left. V_{t+1}(k_{t+1}, z_{t+1}, x_{t+1}) + d V_{t+1}^d(k_{t+1}, x_{t+1}) \right\}$$

$$V_t^{fd}(k_t, z_t, x_t) = \max_{c_t, k_{t+1}} \{ u(c_t, 0) + \beta \sum_{x_{t+1}} \psi(x_{t+1} | x_t)$$

$$\left[ d V_t^d(k_{t+1}, x_{t+1}) + (1-d) \sum_{z_{t+1}} \pi(z_{t+1} | z_t) \right.$$

$$\left. V_{t+1}(k_{t+1}, x_{t+1}, z_{t+1}) \right\}$$